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Summary

Dynamic Software Engineer with robust experience in designing and implementing secure, scalable server-side solutions using Node.js, Microservices architecture, and advanced API design. Complementing my backend expertise is a strong foundation in front-end development with React.js, Next.js, and Tailwind CSS, allowing me to create intuitive and responsive user interfaces. Skilled in cloud computing with AWS and containerization with Docker, I integrate cutting-edge technologies such as AI-powered systems and secure payment gateways to build holistic, high-performance applications. Known for delivering high-quality code and collaborating effectively with cross-functional teams, I continuously seek to enhance my technical expertise across the full development stack.

Soft Skills

- Communication Skills
- Problem-Solving:
- Time Management
- Teamwork & Collaboration:
- Adaptability
- Creativity & Innovation
- Team Leader

Technical Skills

Front-End Development:

1. React.js:

- **Component-based architecture:** Building reusable UI components for modular development.
- **State management:** Experience with **hooks** (useState, useEffect) and **context API** for managing global state, with advanced knowledge of **Redux** for scalable state management in complex applications.
- **Performance optimization:** Techniques like lazy loading, code splitting, and memoization to improve app performance.

2. Next.js:

- **Server-side rendering (SSR)** and **static site generation (SSG)** for improved performance and SEO.
 - **API routes**: Building back-end API endpoints within a Next.js framework.
 - **Advanced routing**: Creating dynamic routes and managing nested layouts.
 - **Services**: Using services for business logic and integrating them with **RESTful APIs**.
3. **Tailwind CSS**:
- **Utility-first CSS framework**: Implementing responsive, mobile-first designs using utility classes.
 - **Custom themes**: Creating and managing consistent themes across multiple applications.
 - **CSS-in-JS**: Familiarity with styled-components and using CSS with JavaScript for dynamic styling.

Back-End Development

- **Node.js**:
 - **MVC Architecture**: Designing scalable applications using Model-View-Controller architecture for clean and maintainable code.
 - **Express.js Framework**: Developing RESTful APIs and managing routing, middleware, and services.
 - **Asynchronous Programming**: Mastering callbacks, promises, and async/await for handling non-blocking operations.
 - **API Security**: Implementing OAuth and JWT for secure authentication and token-based authorization.
- **Microservices Architecture**:
 - **Service-Oriented Design**: Built and deployed microservices-based applications using Node.js, Express, and MongoDB.
 - **Event-Driven Architecture**: Used RabbitMQ for service-to-service communication.
 - **Data Management**: Implemented MongoDB, PostgreSQL, and Redis for distributed data handling and caching.
 - **Containerization & Orchestration**:
 - Deployed services using Docker and managed them with Kubernetes.
 - Utilized Helm charts for defining and packaging microservices configurations.
 - **API Gateway & Load Balancing**: Used Express Gateway to route requests, enforce security, and optimize performance.
- **Advanced Node Concepts**:
 - In-depth understanding of the Node Event Loop, Libuv thread pool, and OS-level operations.

- Expertise in clustering, forking child processes, and using worker threads (including PM2 and webworker threads) for performance optimization.
 - Detailed knowledge of Node internals—from module implementations to unexpected event loop behaviors.
- **Enhanced Performance Tuning:**
 - ▢ Focused on addressing event loop blocking, practical clustering, forking, benchmarking server performance, and PM2 configuration.
- **Node Internals Deep Dive:**
 - ▢ Covered topics include Node module implementations, event loop ticks, C-backed Node operations, and Libuv thread pools.
- **Testing and CI/CD:**
 - Advanced automated headless browser testing with Puppeteer and Jest—including session management, OAuth flow assertions, proxies, and the creation of advanced test helpers.
 - Integration of CI pipelines using Travis CI, YAML configuration, and automated build/test setups.
- **Continuous Integration & Scalable File Uploads:**
 - ▢ Handle scalable image uploads with AWS S3, presigned URL security, and CORS error handling.
- **Data Caching:**
 - Proficient in implementing caching layers with Redis to optimize MongoDB query performance, including handling cache keys, automated cache clearing, and model hydration.
 - **Enhanced Data Caching Techniques:**
 - ▢ Detailed exploration of query caching layers, hash operations, middleware for cache clearing, hydration of models/arrays, toggleable caching mechanisms, and cache expiration management.
- **Database Management:**
 - **SQL Databases:** Strong knowledge of relational databases like MySQL, writing complex queries, optimizing joins, and designing normalized database schemas.
 - **NoSQL Databases:** Experience with MongoDB for handling unstructured data and flexible schemas.

API Development & Integration

- **RESTful APIs:**
 - Designing and developing APIs for efficient data transfer between frontend and back-end systems.

- Ensuring scalability, security, and performance in API-driven architectures.
 - **GraphQL:**
 - Familiar with defining queries, mutations, and managing resolvers for efficient client-server communication.
 - **Payment Gateways:**
 - Integrated secure payment gateways such as Stripe for processing online transactions.
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DevOps & Infrastructure

- **Containerization:**
 - **Docker:** Knowledge of containerizing applications to ensure consistency across environments, managing containers, and Docker images.
 - **Kubernetes:** Experience in orchestrating containerized applications, managing clusters, deploying scalable applications, handling rolling updates, and managing services and networking within Kubernetes.
 - **Cloud Services:**
 - **AWS:** Experience with AWS S3 for storage, EC2 for hosting, Lambda for serverless computing, and DynamoDB for NoSQL database management.
 - **Docker:** Proficient in containerization and deploying applications using Docker.
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API Security & Authentication

- **OAuth:**
 - Implementing OAuth2 for secure user authentication and integrating with third-party services like Google or Facebook for single sign-on (SSO).
- **JWT (JSON Web Tokens):**
 - Handling user sessions and securing APIs using JWT for token-based authentication.
- **Security Best Practices:**
 - Securing web applications using techniques like CORS configuration, HTTPS, and other industry-standard practices.

Professional Experience

Zaio Internship

Position: MERN Stack Developer (Internship)

Duration: March 2023 - March 2024

- Developed and maintained full-stack web applications using **MongoDB, Express.js, React, and Node.js (MERN Stack)**.

- Built and consumed **RESTful APIs** to facilitate seamless communication between the front-end and back-end.
- Implemented **user authentication and authorization** using JWT and OAuth for secure access control.
- Designed and optimized **MongoDB schemas** for efficient data storage and retrieval.
- Created **responsive and dynamic UI components** using React and Tailwind CSS to enhance user experience.
- Integrated third-party services and APIs to extend application functionality.
- Debugged and resolved front-end and back-end issues to ensure smooth application performance.
- Collaborated with senior developers to follow best practices in **code structure, version control (Git), and deployment strategies**.

Aflumed Health care

Position: Junior Software Developer

Duration: May 2024 – Current

- **Advanced Testing & CI/CD:**
 - Pioneered integration of advanced automated testing frameworks using Jest, Puppeteer, and custom test factories to ensure robust, end-to-end quality across both frontend and backend layers.
 - Established CI/CD pipelines with GitHub Actions, Jenkins, and GitLab CI/CD, automating builds, tests, and deployments for continuous delivery.
- **Frontend Excellence:**
 - Developed scalable, high-performance web applications using React.js and Next.js, emphasizing server-side rendering for improved load times and SEO performance.
 - Engineered reusable, responsive UI components with Tailwind CSS, maintaining adherence to modern design principles.
 - Implemented advanced state management with Redux Toolkit to streamline data flow in dynamic applications.
- **Backend & API Innovations:**
 - Architected secure and efficient RESTful APIs and GraphQL endpoints, leveraging OAuth2 and JWT-based authentication for safe data transactions.
 - Applied deep Node.js knowledge including performance tuning, Redis caching for optimized query handling, and scalable file handling with AWS S3.
 - Employed advanced Node internals concepts such as the event loop, worker threads, and clustering to maximize application performance and reliability.
- **Cross-Functional Collaboration:**
 - Coordinated closely with cross-functional teams to integrate advanced backend services with responsive frontend interfaces.

Projects

Project Name: Learning Management System (LMS)

Project Link : [LMS](#)

AWS Architecture : [LMS AWS Architecture](#)

DynamoDB Design : [LSM DynamoDb Design](#)

Project Description:

A full-featured learning management system that allows users to create and manage courses, upload images and videos, and share educational content. Students can purchase courses via Stripe for secure payment processing. The platform also includes features like course organization, student progress tracking, multimedia-rich lessons, and secure user authentication.

Front-End Tools Used:

- Next.js: A React framework for building server-side rendered and static web applications with optimized performance and SEO capabilities.
- Redux Toolkit: A state management library for React that simplifies the management of global state and actions.
- Clerk: A modern authentication and user management service for secure login, registration, and session handling.
- Vercel: A cloud platform for hosting front-end applications with automatic deployments, continuous integration, and optimal performance.

Backend Tools Used:

- Node.js: A JavaScript runtime for building scalable and fast server-side applications.
- Express: A minimalist web framework for Node.js that simplifies routing and middleware management.
- AWS S3: A scalable cloud storage service for storing images, videos, and other static assets.
- AWS API Gateway: A service for creating, publishing, and managing RESTful APIs to handle HTTP requests.
- AWS Lambda: A serverless compute service that runs backend code in response to HTTP requests or events.
- AWS DynamoDB: A fully managed NoSQL database service for storing and retrieving application data.

- AWS CloudFront: A content delivery network (CDN) for fast distribution of static content and media to users globally.
- AWS ECR: A fully managed container registry for storing and managing Docker container images.
- Docker: A platform for containerizing applications, ensuring consistent environments across development, testing, and production.
- Stripe: A payment processing platform that enables secure transactions for course purchases

Project Name: Project Management Dashboard

Project Link : [Project Management](#)

AWS Architecture : [PMD AWS Architecture](#)

Amazon RDS Design : [PMD RDS Design](#)

Project Description:

Built a comprehensive Project Management Dashboard that enables users to manage projects, track timelines, and organize tasks. The platform supports real-time collaboration with features like team management, project creation, task prioritization, and a Gantt chart for visualizing timelines. The dashboard integrates with Clerk for user authentication and leverages AWS EC2 for backend hosting. The app also provides an intuitive interface for managing data with features like project search, team assignment, and user management.

Front-End Tools:

- **Next.js:** A React framework for building server-side rendered and static web applications with enhanced performance.
- **Redux Toolkit:** A library for managing application state and simplifying data flow across components.
- **Redux Toolkit Query:** A tool for fetching and caching data, reducing the need for repetitive API calls.
- **Clerk:** A service for user authentication, providing sign-up, sign-in, and access control features.
- **Material UI Data Grid:** A powerful data grid component for displaying complex, structured data with sorting and filtering capabilities.

Backend Tools:

- **Node.js:** A JavaScript runtime for building scalable, server-side applications.
- **Express:** A minimal web framework for Node.js that simplifies API route creation and middleware handling.

- **PostgreSQL:** A relational database system used to store and manage project data.
 - **Prisma:** A next-gen ORM for Node.js and TypeScript that simplifies database queries and migrations.
 - **AWS EC2:** A cloud computing service for deploying and managing virtual servers.
 - **AWS RDS:** A managed relational database service for storing PostgreSQL databases.
 - **AWS API Gateway:** A service to create, manage, and monitor APIs for backend communication.
 - **AWS S3:** A storage service for hosting files, including images and documents.
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Education

National Certificate: Information Technology: Systems Development

Issued by: MICT SETA

Date of Issue: 30 September 2024

NQF Level: 5

Diploma in Business Information Technology

Issued by: University of Johannesburg

Year : 2021 - 2023

NQF Level: 6

Certification

AWS Certified Developer Associate(in progress)

AWS Cloud Practitioner
